Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Lesson 1: Sum Rule, Product Rule, and Bayes' Theorem

- Sum Rule, Product Rule, and Bayes' Theorem — Reading • 15 min
- Sum Rule — Reading • 4 min
- Product Rule (Chain Rule) — Reading • 2 min
- Bayes' Theorem — Reading • 43 min

Lesson 2: Gaussian Distribution

- Univariate Gaussian Distribution — Reading • 4 min
- Gaussian Distribution: Foundations and Applications — Reading • 15 min
- Multivariate Gaussian Distribution — Reading • 10 min
- Conditional and Marginal Multivariate Gaussian Distributions — Reading • 45 min
- Product of Gaussian Densities — Reading • 26 min

Lesson 3: Inference Techniques

- **Bayesian Inference** — Reading • 1h 10m
- Latent-Variable Models — Reading • 50 min
- Check Your Knowledge: Inference Techniques — Graded Assignment • 10 min

Module 4
Introduction to Model Evaluation and Optimization

Bayesian Inference

0:45 / 3:00 2x

Introduction to Bayesian Inference

- Bayesian inference is a key method in machine learning for estimating hidden variables θ from observed data x.
- Unlike maximum likelihood (ML) or maximum a posteriori (MAP) estimation, which yield single point estimates, Bayesian inference derives a full posterior distribution of parameters.

Importance of the Posterior Distribution

- Focusing solely on point estimates (θ^*) can lead to a loss of information critical for decision-making.
- Bayesian inference finds the posterior distribution p(θ|X)using Bayes' theorem, providing a more robust foundation for predictions and decision-making by incorporating the full uncertainty in parameter estimates.

Bayesian Inference Mechanics

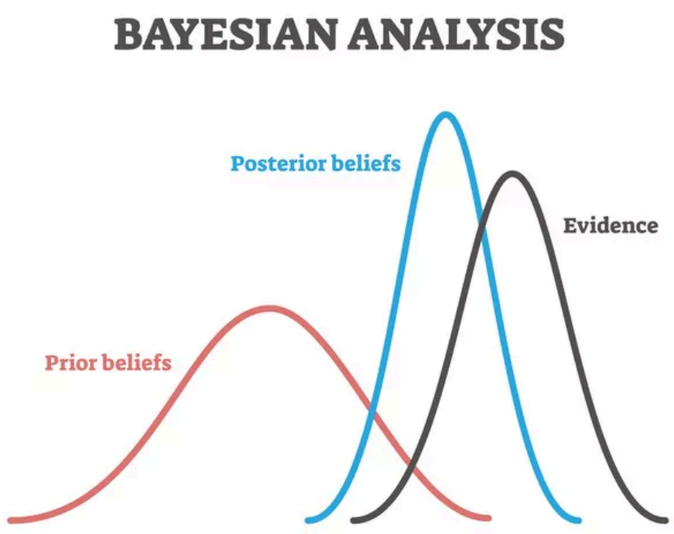
Bayes' Theorem:

For a dataset X , a parameter prior $p(\theta)$, and a likelihood function, the posterior

$$p(\boldsymbol{\theta} \mid \mathcal{X}) = \frac{p(\mathcal{X}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{X})}, \quad p(\mathcal{X}) = \int p(\mathcal{X} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})\mathrm{d}\boldsymbol{\theta}$$

Prediction: The predictive distribution p(θ) on the parameters

$$p(\boldsymbol{x}) = \int p(\boldsymbol{x} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})\mathrm{d}\boldsymbol{\theta} = \mathbb{E}_{\boldsymbol{\theta}}[p(\boldsymbol{x} \mid \boldsymbol{\theta})]$$



Comparison with Parameter Estimation

- Parameter Estimation: Provides a single best estimate $(\theta*)$ by solving an optimization problem.
- Bayesian Inference: Provides a posterior distribution by solving an integration problem, allowing for better handling of uncertainty and incorporation of prior knowledge.

Applications of Bayesian Inference

- Applied in diverse fields such as large-scale topic modeling, click-through-rate prediction, reinforcement learning, online ranking systems, and recommender systems.
- Tools like Bayesian optimization are crucial for efficient hyperparameter tuning in models.

Additional Resources:

Bayesian Inference Interactive - Brown University (15 minutes)

This interactive, created by Brown University gives an interactive, highly visual explanation of Bayes Theorem and Bayesian Inference.

Fundamentals of Statistics: Bayesian Inference - Marco Taboga (15 minutes)

This is an online written resource by Marco Taboga that gives a critical look into the in-depth mathematics behind Bayesian Inference and its importance to statistics.

Bayesian Inference in Machine Learning - Wolfram (30 minutes)

This article from Wolfram's knowledge base goes into great detail on Bayesian Inference in machine learning, including practical knowledge on this key application of the discipline.

Video:

Super Simple Explanation of Bayesian Statistics for Dummies - Quant Psych (5 minutes)

This short video by Quant Psych breaks down Bayesian Inference in a quick and useful way. Though it is "for dummies," it's still a great way to get into the topic.

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